

Cryptographic posture assessment

Synthetic lab PCAP

lab-expired-certificate.pcap

Capture timestamp: 2026-10-04T12:00:00+00:00

Source SHA-256: a5fe45457b23f320ff415382f0782026e6b9f28a88a838fdce6c372984d99a28

Generated: 2026-10-05T13:52:42Z

MAIL SESSIONS	POSTURE	EVIDENCE GAPS
1	HIGH_RISK	0

SecureMailScope analyzed 1 observed SMTP transport session(s) from capture 'lab-expired-certificate.pcap'. 0 session(s) passed all assessed baseline checks, 0 required review, and 1 had critical transport risks. Significant transport weaknesses or obsolete cryptographic protocols were identified. The assessment identified 3 finding(s) across transport sessions. AI-assisted anomaly prioritization was not performed because only 1 observed client(s) was available for comparison (minimum requirement: 5 observed clients).

Prioritized security findings

HIGH / FAIL / SMS-CERT-001

Certificate assessment: issues detected

Expired at capture time; Self-issued leaf certificate; explicit trust required

Evidence: Expired at capture time; Self-issued leaf certificate; explicit trust required

Recommendation: Renew invalid certificates, use modern keys, and supply an approved PEM trust store via SMS_TRUST_STORE for offline path validation. Check revocation separately.

References: RFC 5280, RFC 8446

[RULE] Recommended fix

1. Check the reported certificate issue against the intended hostname and capture time.
2. Renew expired certificates, correct names and intermediates, and use approved keys and signature algorithms as applicable.
3. Supply an approved PEM trust bundle for offline path validation. Assess revocation using separately obtained evidence; this tool makes no revocation requests.

INFO / PASS / SMS-TLSVER-002

Modern TLS Version (TLS 1.2)

A currently accepted TLS protocol version (TLS 1.2) was observed.

Evidence: Server negotiated modern protocol version TLS 1.2.

Recommendation: Maintain current TLS version baseline and monitor for future cryptographic standard updates.

References: RFC 5246, NIST SP 800-52 Rev. 2

[RULE] Recommended fix

1. No change is required by this observed check. Maintain the baseline and re-check a fresh capture after configuration changes.

INFO / PASS / SMS-PFS-001

Forward Secrecy Supported

The negotiated cipher suite utilizes ephemeral key exchange (ECDHE/DHE), providing Perfect Forward Secrecy.

Evidence: Negotiated cipher suite TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256 includes ephemeral key exchange.

Recommendation: Continue enforcing ephemeral key exchanges (ECDHE) across all mail transport connections.

References: RFC 8446, NIST SP 800-52 Rev. 2

[RULE] Recommended fix

1. No change is required by this observed check. Maintain the baseline and re-check a fresh capture after configuration changes.

Session evidence and certificates

SMTP-STREAM-0: Client (192.0.2.1) -> Server (198.51.100.25)

SMTP / Direct TLS (Implicit) / TLS 1.2 / TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256

ServerHello frame: 6. Packets: 15. STARTTLS requested: False; accepted: False; ServerHello after upgrade: False.

INFO / [COVERAGE_GAP] Quantum readiness: UNKNOWN

Group: UNKNOWN. A recognized TLS 1.3 ServerHello key-share group is not visible. No readiness conclusion or failing check is produced.

CN=mail.sih.test

Issuer: CN=mail.sih.test. Key: RSA 2048. Signature hash: sha256.

Validity: 2026-06-26T00:00:00+00:00 to 2026-10-03T00:00:00+00:00. Capture frame: 6.

No explicit trust store supplied. Chain trust and revocation are not verified.

Certificate SHA-256: 513a590c0990293da2d8a6b4dafa8828177a78775d69976c6bf2f917edc9d22

AI-assisted prioritization

Only 1 observed client is present in this capture. Statistical outlier analysis requires a larger comparison population (minimum 5 observed clients).

AI anomaly scores provide statistical prioritization relative to the observed capture fleet only. AI assessments never declare maliciousness or replace deterministic Phase 2 security findings.

Assessment boundaries

- TLS selections come from ServerHello; observing ServerHello does not prove handshake completion or successful delivery.
- TLS 1.3 certificates are encrypted. Chain validation requires visible certificates, observed SNI, capture time, and SMS_TRUST_STORE. Revocation is not checked.
- Capture SHA-256 and report hashes support integrity comparison, not source authentication or digital signatures.
- Analysis reflects strictly traffic captured and visible within the submitted packet capture window.
- Passive network observation cannot evaluate email clients or cryptographic capabilities absent from the capture.
- Incomplete TCP streams or truncated TLS handshakes in capture boundaries may limit full capability assessment.
- AI-assisted anomaly prioritization is relative strictly to the observed client population in this capture.
- AI anomaly engine abstains from statistical ranking when the comparison population is insufficient (minimum 5 observed clients).
- Encrypted application-layer email payload (message text, headers, attachments) is not decrypted.
- Hardening Impact Simulation results reflect passive capability evidence only and do not modify live servers.

Evidence integrity

Report digest: a0b813c4e751c07e2d3ba48f6f7f0d5173db9cd2bc76bd162eb9e38d8777b4e5

Hashes detect changes only when compared with an independently retained reference. They are not digital signatures and do not authenticate the analyst or the capture origin.

Capture files can contain sensitive payloads and are processed locally by TShark. Temporary uploaded files are removed after analysis. Command arguments are discarded at the extraction boundary; generated investigations retain transport and certificate metadata rather than message bodies or authentication secrets. Reports may contain IP addresses, hostnames, and certificate identities. No network revocation lookup or cloud AI is used.